# From a five-torsion screen to genus-three existence

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This is a derived sufficient genus-three existence criterion. It extends the module theorem in `CUBIC_GLUING.md`; it does not reconstruct a curve or establish publication novelty. The exact check is frozen separately in `FIVE_GLUING_GEOMETRY_CHECKS.md`.

## Indecomposability lemma

Let $A = E_0 \times E_1 \times E_2$ with its product principal polarization over an algebraically closed field of characteristic different from five. Let $L \subset A[5]$ be a maximal isotropic subgroup transverse to each coordinate elliptic curve. Suppose no geometric isogeny of degree at most six joins distinct $E_i$. Then the principally polarized quotient $B = A/L$ is geometrically indecomposable.

**Proof.** Write $\phi : A \to B$ for the quotient. If B were decomposable, a principally polarized threefold has a principally polarized elliptic factor D. Restrict the polarized dual $\phi^\dagger : B \to A$ to D, and write its components $f_i : D \to E_i$. Since $(\phi^\dagger)^* \lambda_A = 5\lambda_B$, the nonzero component degrees $d_i = \deg f_i$ sum to five. If two components are nonzero, composing $f_j \widehat{f_i} : E_i \to E_j$ gives an isogeny of degree $d_i d_j \leq \lfloor 25/4 \rfloor = 6$, a contradiction. If only one component is nonzero, its degree is five. The identity $\phi\phi^\dagger = [5]$ on D then implies that $\phi|_{E_i}$ has degree five as well, since $\deg [5]_D = 25$. This contradicts $L \cap E_i = 0$. Thus B is indecomposable.

The degree-six hypothesis is sufficient, not asserted necessary. This is the same polarization-degree argument as the earlier three-gluing lemma, with the extra degree partitions accounted for explicitly.

## Binary cubic criterion

Let $K = \mathbb{F}_{2^{3d}}$, $k = \mathbb{F}_{2^d}$, $d \geq 3$, and $E_a/K : y^2 + xy = x^3 + ax^2 + b$, with b of full absolute degree $3d$. Let t be its K-trace and $q = 2^d$. Assume:

1. $t^2 - 4q^3$ is a nonzero square modulo five.
2. For $j = 1/b$ and $y = j^q$, all four values $\Phi_3(j, y), \Phi_5(j, y), \Phi_3(j^2, y), \Phi_3(j, y^2)$ are nonzero.

Then one of E and its nontrivial quadratic twist admits, through its Weil restriction, a k-rational $(5, 5, 5)$-isogeny to the Jacobian of a smooth genus-three curve over k. The isogeny is injective on every prime subgroup of order $r \neq 5$ of that selected source twist.

**Proof.** The discriminant hypothesis gives two distinct rational eigenvalues on E[5]; apply the module theorem to obtain L. All source conjugates are ordinary and have full absolute j-degree. Over an algebraic closure an ordinary binary elliptic curve's degree-two isogeny targets have j-invariants $j^2$ and $j^{1/2}$. Degree-four targets have invariants $j^4, j, j^{1/4}$. Full degree and $d \geq 3$ exclude equality with a distinct relative conjugate, so degrees 1,2,4 cannot occur.

The classical modular polynomials exclude degrees 3 and 5. A degree-six isogeny factors into degrees two and three. Its j-pair therefore satisfies $\Phi_3(j^2, y) = 0$ or $\Phi_3(j^{1/2}, y) = 0$. Squaring the latter equation is equivalent to $\Phi_3(j, y^2) = 0$. The four tests exclude these cases. Checking the pair $(j, j^q)$ covers all conjugate pairs: Frobenius cycles the indices, and duality reverses an isogeny without changing its degree. Thus the lemma applies.

An indecomposable principally polarized threefold over an algebraically closed field is a genus-three Jacobian. Arithmetic Torelli over the finite base field gives either B or its [-1]-twist as a Jacobian over k. The odd relative degree three makes this twist the Weil restriction of the source's nontrivial quadratic twist. The degree of the gluing isogeny is $5^3$; its kernel cannot meet an r-subgroup for $r \neq 5$.

This proof does not choose which twist works, assert hyperellipticity, construct C, evaluate the isogeny or bound the total attack cost. These distinctions are required before promoting a candidate into an attack catalogue. The general theorem and geometric bridges have not been formalized in Lean; the pinned fixed module proof is partial.

## Exact modular polynomials and sources

The retained integer coefficients from Sutherland's MIT distribution reduce in characteristic two to

$$\Phi_3(X,Y) = X^4 + Y^4 + X^3Y^3,$$
$$\Phi_5(X,Y) = X^6 + Y^6 + X^5Y^5 + X^4Y^2 + X^2Y^4.$$

`cubic-gluing-reference/PROVENANCE.json` binds their source URLs and SHA-256 digests. Rust unit checks parse the retained integer coefficients and compare the complete reductions, including symmetry. This verifies the implementation against those fixtures, not a new computation of the classical modular polynomials. The distribution cites Bröker–Lauter–Sutherland, 2010 for prime-level computation.

The Jacobian existence and arithmetic-twist steps use the same published results and limits as `THREE_GLUING.md`: Oort–Ueno and Beauville–Ritzenthaler, with the finite-field Torelli distinction preserved. Tian's preprint already uses the isotropic quotient approach for cubic cover attacks. Hess's broad binary descent inclusion remains a baseline. Exact recognition priority and the full 2023 journal comparison are unresolved.

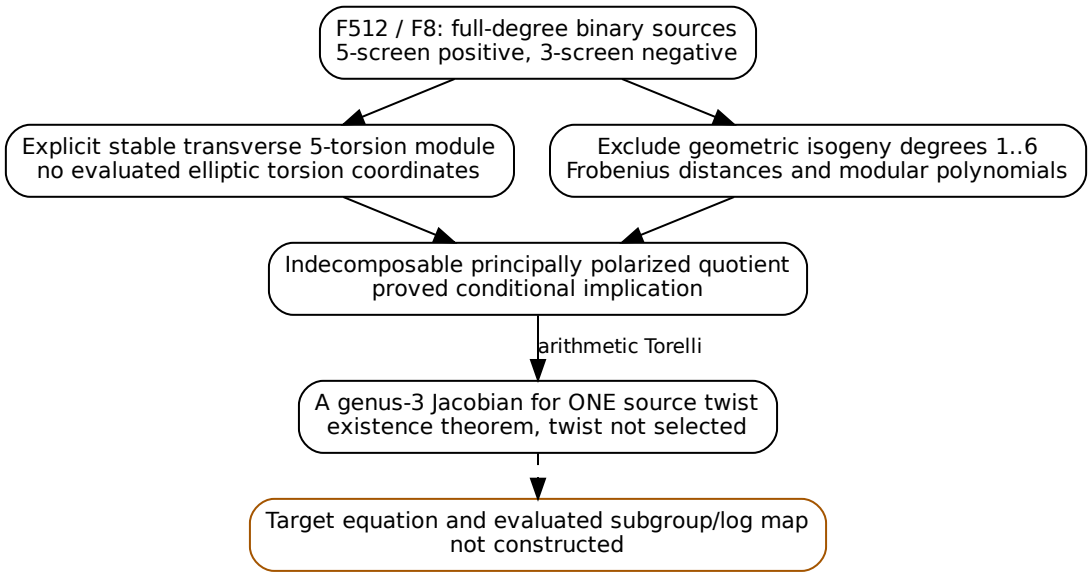


Figure 1: The solid part ends at existence for one source twist. Target reconstruction and a log transfer remain open.

The previous module diagram is not rewritten; this diagram adds the new geometric hypotheses. No canonical curve or isogeny route is added without the explicit selected target and checked map. Results are recorded in the new dated `CUBIC_GLUING_RESULTS.md`, preserving the preregistered notes.